

Therapeutic Landscape Snapshot:
Comorbid Depression in Idiopathic Generalized Epilepsy and
the Parallel Development of Azetukalner

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Idiopathic generalized epilepsy (IGE) accounts for 15%–20% of all epilepsies.¹ Approximately 20%–30% of patients with epilepsy experience comorbid depression.^{2,3} Previous questionnaire research has reported that patients with IGE have mood dysfunction and poor quality of life.⁴⁻⁷ In patients with mesial temporal lobe epilepsy, concurrent mood and anxiety disorders have also been associated with pharmacoresistant seizures.⁸

Valproate is a primary medication for seizure control, and approximately 85% of patients with juvenile myoclonic epilepsy achieve good seizure control on it.⁹ However, valproate carries a high risk of teratogenicity. In the United Kingdom, the Medicines and Healthcare products Regulatory Agency (MHRA) requires that valproate not be started in patients of either sex younger than 55 years unless 2 specialists independently document that no other treatment is effective or tolerated. Women and girls of childbearing potential who take valproate must also be enrolled in a Pregnancy Prevention Programme (PPP) and review their risks annually with a specialist.¹⁰ Notably, most patients with IGE have an earlier onset age, in childhood or adolescence, so the restricted population and the treatable population substantially overlap. Levetiracetam is therefore a common alternative for patients who cannot use valproate.⁹ However, levetiracetam has been associated with behavioral adverse events, with incidence rates of 9.9% for irritability, 2.5% for anger, and 2.6% for aggression.¹¹

Previous research has indicated a bidirectional relationship between seizures and depression.¹² Among patients with newly diagnosed unprovoked seizures, a history of major depression was 1.7-fold more common than among controls (95% CI, 1.1–2.7), and a history of attempted suicide was 5.1-fold more common (95% CI, 2.2–11.5). In a randomized controlled trial of the Kv7 channel opener ezogabine in major depressive disorder (MDD), the primary neuroimaging endpoint was not met; however, depression severity on the Montgomery-Åsberg Depression Rating Scale (MADRS) and anhedonia on the Snaith-Hamilton Pleasure Scale (SHAPS) improved

significantly compared with placebo.¹³ Azetukalner, currently in clinical trials, also targets Kv7.2/Kv7.3 potassium channels.¹⁴

Although previous research has documented the comorbidity between epilepsy and depression, no azetukalner trial has assessed both seizure and mood outcomes in the same population. Seizure outcomes and mood outcomes are being studied in separate trials (epilepsy: NCT03796962, NCT05614063, NCT05716100, NCT05667142, NCT05718817; MDD: NCT05376150, NCT04827901, NCT06775379, NCT07076407, NCT06922110; bipolar depression: NCT07172516, NCT07217860). X-NOVA was a phase 2 proof-of-concept trial of azetukalner in adults with MDD. Its primary endpoint, change in MADRS score at week 6, was not met significance: the mean (SE) change was -16.94 (1.45) points with 20 mg azetukalner and -13.90 (1.41) points with placebo (difference, -3.04 points; 95% CI, -7.04 to 0.96; $p = 0.140$). The secondary anhedonia endpoint, change in SHAPS score, showed a significantly greater reduction with 20 mg azetukalner than with placebo (-7.77 (0.87) vs -5.30 (0.85) points; $p = 0.046$), as did the exploratory endpoint of change in 17-item Hamilton Depression Rating Scale (HAM-D17) score (-13.3 (1.1) vs -10.2 (1.0) points; $p = 0.040$).¹⁵

By contrast, X-ACT, the phase 3 trial of azetukalner in primary generalized tonic-clonic seizures (PGTCS), enrolls patients aged 12 years or older who have not achieved sustained seizure freedom with at least 2 antiseizure medications (ASMs). Its primary endpoint is median percent change in monthly seizure frequency, and its secondary endpoints are the 50% responder rate and seizure freedom; no mood or quality-of-life endpoint is included. The trial also excludes patients with bipolar disorder and those with uncontrolled unipolar major depression requiring changes in pharmacotherapy (NCT05667142). Comorbid depression can affect both seizure response to ASMs and adverse events, potentially biasing the interpretation of epilepsy trial findings.¹⁶ While this justifies excluding patients with uncontrolled depression or bipolar disorder from trials such as X-ACT, it remains unclear how drugs developed for both conditions affects patients who have both.

Implications

Retrospective chart review and real-world data

Although the azetukalner epilepsy trials do not measure mood, retrospective chart review and real-world data allow us to investigate patients' mood states under existing ASMs and their relationship with depressive symptoms. Depression in people with epilepsy is clinically heterogeneous, and whether its presentation differs from that of primary depression remains debated.¹⁶ The Neurological Disorders

Depression Inventory for Epilepsy (NDDI-E) was developed to identify depressive symptoms that do not overlap with cognitive deficits or ASM adverse effects; its association with major depression is not affected by ASM adverse effects, whereas that of generic screening instruments is.¹⁷ Because the NDDI-E has been validated in outpatient epilepsy clinic settings,¹⁸ relevant screening data may already exist in clinical records.

Insight from ezogabine's history

Ezogabine was the first clinically Kv7.2/Kv7.3 activator approved for the treatment of focal-onset seizures. It was withdrawn from the market following declining use after the discovery of pigmentation changes in the retina and skin. Although early clinical studies of azetukalner have shown no evidence of pigmentation-related adverse effects, continued monitoring of effects remains warranted.¹⁴

Timing of the azetukalner readouts

The primary completion date for the first azetukalner phase 3 study in MDD (X-NOVA2, NCT06775379) is listed as August 2026; the primary completion date for X-ACKT is April 2027. Accordingly, data in MDD are expected to reach neurologists before any epilepsy-specific data are available. X-ACKT will not answer how azetukalner affects mood in patients with epilepsy, because mood is not its endpoints. Therefore, we recommend that the retrospective chart review be completed before the end of this year. A position will be required regardless of the X-ACKT result.

Prepared as a writing sample. All information is drawn from published literature and public trial registries; no unpublished or client data are included. Reference 6 is the author's own first-author publication.

References

1. Jallon P, Latour P. Epidemiology of idiopathic generalized epilepsies. *Epilepsia*. 2005;46 Suppl 9:10-14. doi:10.1111/j.1528-1167.2005.00309.x
2. Fiest KM, Dykeman J, Patten SB, et al. Depression in epilepsy: a systematic review and meta-analysis. *Neurology*. 2013;80(6):590-599. doi:10.1212/WNL.0b013e31827b1ae0
3. Scott AJ, Sharpe L, Hunt C, Gandy M. Anxiety and depressive disorders in people with epilepsy: A meta-analysis. *Epilepsia*. 2017;58(6):973-982. doi:10.1111/epi.13769

4. Shehata GA, Bateh Ael-A. Cognitive function, mood, behavioral aspects, and personality traits of adult males with idiopathic epilepsy. *Epilepsy Behav.* 2009;14(1):121-124. doi:10.1016/j.yebeh.2008.08.014
5. Bolden LB, Griffis JC, Nenert R, Allendorfer JB, Szaflarski JP. Cortical excitability affects mood state in patients with idiopathic generalized epilepsies (IGEs). *Epilepsy Behav.* 2019;90:84-89. doi:10.1016/j.yebeh.2018.11.001
6. Lien Y, Sharma AA, Willoughby WR, et al. Cerebral blood flow increases in idiopathic generalized epilepsy and its relationship with mood dysfunction. *Epilepsia Open.* 2025;10(3):906-917. doi:10.1002/epi4.70049
7. Tracy JL, Dechant V, Sperling MR, Cho R, Glosser D. The association of mood with quality of life ratings in epilepsy. *Neurology.* 2007;68(14):1101-1107. doi:10.1212/01.wnl.0000242582.83632.73
8. Nogueira MH, Yasuda CL, Coan AC, Kanner AM, Cendes F. Concurrent mood and anxiety disorders are associated with pharmacoresistant seizures in patients with MTLE. *Epilepsia.* 2017;58(7):1268-1276. doi:10.1111/epi.13781
9. Serafini A, Gerard E, Genton P, Crespel A, Gelisse P. Treatment of Juvenile Myoclonic Epilepsy in Patients of Child-Bearing Potential. *CNS Drugs.* 2019;33(3):195-208. doi:10.1007/s40263-018-00602-2
10. Medicines and Healthcare products Regulatory Agency. Valproate (Belvo, Convulex, Depakote, Dyzantil, Epilim, Epilim Chrono or Chronosphere, Episenta, Epival, and Syonell▼): review by two specialists is required for initiating valproate but not for male patients already taking valproate. GOV.UK. Published February 13, 2025. Accessed September 10, 2026. <https://www.gov.uk/drug-safety-update/valproate-belvo-convulex-depakote-dyzantil-epilim-epilim-chrono-or-chronosphere-episenta-epival-and-syonellv-review-by-two-specialists-is-required-for-initiating-valproate-but-not-for-male-patients-already-taking-valproate>
11. Steinhoff BJ, Klein P, Klitgaard H, et al. Behavioral adverse events with brivaracetam, levetiracetam, perampanel, and topiramate: A systematic review. *Epilepsy Behav.* 2021;118:107939. doi:10.1016/j.yebeh.2021.107939
12. Hesdorffer DC, Hauser WA, Olafsson E, Ludvigsson P, Kjartansson O. Depression and suicide attempt as risk factors for incident unprovoked seizures. *Ann Neurol.* 2006;59(1):35-41. doi:10.1002/ana.20685
13. Costi S, Morris LS, Kirkwood KA, et al. Impact of the KCNQ2/3 Channel Opener Ezogabine on Reward Circuit Activity and Clinical Symptoms in Depression: Results From a Randomized Controlled Trial. *Am J Psychiatry.* 2021;178(5):437-446. doi:10.1176/appi.ajp.2020.20050653
14. Perucca E, Tagliatela M. Targeting Kv7 Potassium Channels for Epilepsy. *CNS Drugs.* 2025;39(3):263-288. doi:10.1007/s40263-024-01155-3

15. Butterfield NN, Luzon Rosenblut C, Fava M, et al. Azetukalner, a Novel KV7 Potassium Channel Opener, in Adults With Major Depressive Disorder: A Randomized Clinical Trial. *JAMA Netw Open*. 2025;8(5):e2514278. Published 2025 May 1. doi:10.1001/jamanetworkopen.2025.14278
16. Kanner AM, Schachter SC, Barry JJ, et al. Depression and epilepsy, pain and psychogenic non-epileptic seizures: clinical and therapeutic perspectives. *Epilepsy Behav*. 2012;24(2):169-181. doi:10.1016/j.yebeh.2012.01.008
17. Gilliam FG, Barry JJ, Hermann BP, Meador KJ, Vahle V, Kanner AM. Rapid detection of major depression in epilepsy: a multicentre study. *Lancet Neurol*. 2006;5(5):399-405. doi:10.1016/S1474-4422(06)70415-X
18. Kim DH, Kim YS, Yang TW, Kwon OY. Optimal cutoff score of the Neurological Disorders Depression Inventory for Epilepsy (NDDI-E) for detecting major depressive disorder: A meta-analysis. *Epilepsy Behav*. 2019;92:61-70. doi:10.1016/j.yebeh.2018.12.006